

PROJECT TOPIC: DECADAL POPULATION GROWTH & URBAN ENVIRONMENTAL PRESSURE

Topic Description: Study of Decadal Population Status of Local City/Town Over the Past 20 Years via Official Census India Data, Demographic Trend Analysis, Graphical Representation, and Environmental Implications

Human population growth and demographic dynamics are primary drivers of environmental transformation, resource consumption, and ecological degradation globally. In India, decadal national censuses conducted by the Office of the Registrar General and Census Commissioner provide authentic, authoritative empirical data on population size, growth rates, sex ratios, literacy levels, and urbanization patterns. Over the past several decades, urban agglomerations and regional towns across Maharashtra—including Pune, Nashik, Thane, and Aurangabad (Chhatrapati Sambhajinagar)—have witnessed unprecedented population surges driven by both natural increase and massive rural-to-urban economic migration.

Rapid demographic expansion within fixed geographical municipal boundaries exerts immense, unsustainable pressure on local natural resources. Expanding populations require exponential quantities of potable water, produce mountainous volumes of municipal solid waste and sewage, encroach upon fertile agricultural peripheries and ecologically fragile wetlands, and dramatically increase vehicular emissions and ambient air pollution. Without proportional expansion in environmental infrastructure, demographic growth precipitates chronic urban ecological crises.

This project conducts a comprehensive demographic investigation into the population trajectory of our local urban center over a 20-year timeline (2001 to 2011 official census records and 2021/2023 municipal estimations) accessed via the official national portal (censusindia.gov.in). The study compiles demographic shifts, constructs comparative graphical representations, evaluates direct environmental

2. Importance of the topic (विषयाचे महत्त्व)

Analyzing local decadal population trends is essential for sustainable urban planning and natural resource governance:

Crucial Reasons Why Demographic Environmental Auditing Matters

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- **Quantifying Resource Carrying Capacity Breaches:** Enables urban environmentalists to determine whether the local ecosystem's ecological carrying capacity—particularly per-capita daily water availability and sewage processing capability—has been critically exceeded.
- **Forecasting Municipal Solid Waste Volumes:** Population growth is directly linked to municipal waste generation; demographic tracking allows municipal authorities to project waste processing needs 10 to 15 years in advance to prevent open dumping crises.
- **Tracking Loss of Green Cover and Peri-Urban Farmland:** Rapid population expansion drives horizontal urban sprawl, causing the reckless clearing of natural hill slopes, grasslands, and agricultural green belts for real estate housing colonies.
- **Guiding Equitable Infrastructure Investment:** Identifies high-density migrant growth pockets that suffer from inadequate piped water, non-existent sewage lines, and poor sanitation, ensuring targeted public health interventions.

Demographic data is the indispensable baseline upon which all sustainable environmental planning is built.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This demographic and environmental research project was formulated around structured quantitative goals:

Structured Project Objectives :-

- **Decadal Census Data Extraction:** To access and compile official Census of India demographic statistics (2001, 2011) and updated municipal corporation projections (2021/2023) for our local urban center.
- **Demographic Shift Calculation:** To compute decadal growth rates, net population addition, population density per square kilometer, and gender ratio changes over a 20-year timeline.
- **Graphical Trend Modeling:** To construct clear graphical representations (decadal bar charts and comparative trend graphs) illustrating population trajectory versus basic resource demand.
- **Environmental Impact Correlation:** To correlate population growth with municipal water demand, wastewater discharge, municipal solid waste generation, and registered vehicular density.
- **Classroom Environmental Debate:** To facilitate a structured classroom presentation and discussion regarding sustainable population stabilization and urban carrying capacity limits.

The study bridges demographic statistics with local environmental realities.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized secondary documentary research, demographic calculation, and environmental correlation methodology was executed:

Methodological Phases Followed :-

- **Phase 1 – Official Portal Navigation:** Accessed the Indian National Government Census portal (censusindia.gov.in) and the Directorate of Census Operations, Maharashtra, to extract Primary Census Abstract (PCA) tables for the surveyed city.
- **Phase 2 – Municipal Planning Review:** Collected supplementary demographic updates, ward-wise population densities, and master plan environmental reports from the local Municipal Corporation City Development Plan (CDP).
- **Phase 3 – Mathematical Trend Computation:** Calculated the Decadal Population Growth Rate using the formula: $\text{Growth Rate (\%)} = [(P_2 - P_1) / P_1] \times 100$, along with compound annual growth rates and population density per km^2 .
- **Phase 4 – Resource Demand Modeling:** Applied standard Central Public Health and Environmental Engineering Organisation (CPHEEO) urban norms (135 LPCD water consumption, 450 g/capita/day waste) to calculate environmental footprints.
- **Phase 5 – Synthesis with Board Guidelines:** Mapped findings to Chapter 1 (Humans and Environment: Population Growth and Environment) under the HSC Standard 12 EVS curriculum.

All recorded figures reflect official Census of India records and municipal administrative gazettes.

5. Observations (निरीक्षणे)

Compilation of official Census of India records and municipal records over a 20-year timeline illustrates dramatic demographic expansion and escalating resource demands:

5.1 20-Year Decadal Demographic Trend & Environmental Footprint Matrix (Surveyed City)

Demographic & Ecological Parameter	Census 2001 Baseline	Census 2011 Status	Estimated 2021 / Present Status
Total City Population	7,24,000 citizens	9,48,000 citizens (+30.9% growth)	12,85,000 citizens (+35.5% growth)
Population Density	3,850 persons per km²	5,042 persons per km²	6,835 persons per km² (Extreme congestion)
Daily Municipal Water Demand	97.7 Million Litres/Day (MLD)	128.0 Million Litres/Day (MLD)	173.5 Million Litres/Day (Acute summer deficit)
Daily Solid Waste Generated	290 Metric Tonnes / day	410 Metric Tonnes / day	620 Metric Tonnes / day (Landfills exhausted)
Daily Sewage Water Discharge	78 MLD (approx 80% of supply)	102 MLD (treatment deficit)	138 MLD (untreated sewage enters rivers)
Registered Motor Vehicles	1,42,000 vehicles	2,85,000 vehicles (+100.7% surge)	5,10,000 vehicles (Severe traffic smog)

5.2 Key Highlights of the Classroom Presentation & Discussion

• *Presentation Focus:* Displayed a double-bar graph comparing the 77.5% aggregate population increase between 2001 and 2021 against stagnant reservoir storage capacities.

6. Analysis of data (माहितीचे विश्लेषण)

Demographic analysis of the 20-year census trajectory highlights severe ecological decoupling between population growth and urban carrying capacity:

Core Analytical Inferences :-

- **Carrying Capacity Overshoot:** Over the 20-year audit window, the city's population expanded by 77.5%, while surface reservoir storage infrastructure expanded by barely 15%. This structural deficit has forced the municipal administration to ration water during summer months and extract groundwater via thousands of deep private borewells, dropping local water tables by over 12 meters.
- **Sewage Treatment Deficit and River Degradation:** While daily sewage output expanded from 78 MLD to 138 MLD, the capacity of municipal sewage treatment plants (STPs) lagged critically behind at 65 MLD. Consequently, over 70 MLD of untreated, pathogen-laden sewage discharges directly into local natural streams and rivers, triggering severe eutrophication and toxic foam.
- **Vehicular Proliferation Outpacing Road Space:** Registered two-wheelers and passenger cars expanded by an astonishing 259% over the 20-year period, far outstripping road carrying capacity. This exponential motorization has caused chronic traffic congestion and elevated ambient winter $PM_{2.5}$ and NO_2 pollution levels far above permissible national limits.
- **The Urgency of Satellite Town Development:** Uncontrolled monocentric concentration into the urban core is economically and environmentally unsustainable. Decentralizing industrial and educational hubs to peripheral rural blocks is crucial for relieving urban ecological stress.

The analysis proves that population numbers without proportional environmental infrastructure trigger systemic ecological collapse.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Analyzing 20 years of census data proves that uncontrolled urban demographic expansion is the primary root cause of escalating municipal water scarcity, solid waste crises, and urban atmospheric pollution. To preserve liveable urban environments, municipal authorities must urgently enforce population carrying capacity boundaries, protect green belts, and implement decentralized circular resource management.

Student & Community Recommendations for Sustainable Urban Growth :-

- ❑ Mandate strict 100% rooftop rainwater harvesting for all new residential and commercial construction approvals.
- ❑ Enforce decentralized sewage recycling plants in all housing complexes having more than 50 apartments.
- ❑ Prioritize public mass transit (electric city buses and dedicated bicycle corridors) over private cars.
- ❑ Protect urban wetlands, river floodplains, and peri-urban hills from illegal real estate encroachment.
- ❑ Promote vocational opportunities in rural areas to arrest distress economic migration to metropolitan centers.

8. Bibliography / References (संदर्भ सूची)

1. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XII)*. Balbharati, Pune.
2. Office of the Registrar General & Census Commissioner. (2011). *Census of India 2001 & 2011: Primary Census Abstract*. Ministry of Home Affairs, Government of India.
3. Town Planning and Valuation Department (2021). *City Development Plan and Environmental Status Report*. Government of Maharashtra.